# 3-D FDTD Framework for Simulating SAR Imagery of Realistic Near Earth Surface Volumes (Soil, Snow, and Vegetation)

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***Abstract*— Synthetic aperture radar (SAR) imaging of near Earth surface natural material volumes (soil, snow, and vegetation) and man-made objects corresponds to an equivalent real-aperture scenario with temporally short and spatially focused pulses. This equivalence holds as long as the imaged scene can be considered static within both fast (chirp duration) and slow (synthetic aperture) times. The electromagnetic backscatter recorded in a properly focused SAR image resolution cell that meets this condition has an amplitude and phase, which represents the coherent sum of the returns from individual scatterers inside the small volume represented by the cell. For some types of scatterers, the backscattering properties can be accurately characterized by analytical expressions. However, there are many interesting scattering phenomena where a purely analytical treatment leads to unrealistic simplifying assumptions. One approach to investigating these phenomena is with numerical methods, such as the finite-difference time-domain (FDTD) method. The FDTD method is computationally expensive, but it can model the complete physical interaction of electromagnetic waves according to Maxwell's equations with arbitrary materials and shapes. This article describes the development of a specialized 3-D FDTD software package capable of simulating the real-aperture equivalent of individual SAR image resolution cells given the spatial distribution of permittivity and conductivity within the cells. We demonstrate the usefulness of our new software tool by first recreating published 2-D simulation results examining the link between soil moisture and InSAR phase and then expanding these results to more realistic 3-D soil volumes.**

***Index Terms*— Finite-difference time-domain (FDTD) method, interferometry, soil moisture, synthetic aperture radar (SAR).**

## I. Introduction

### A. SAR Image Statistics

THE amplitude and phase of a 3-D resolution cell in a stationary, properly focused synthetic aperture radar (SAR) image depend on the position, orientation, and material properties of the scatterers only inside this cell. Some types of scatterers, such as corner reflectors, have a deterministic response, which can be expressed using closed-form expressions [1]. Other types of scatterers, such as the Amazon rainforest, are known as fully distributed scatterers and can be described statistically [2]. However, there are many other situations in which there are neither analytical nor exact statistical descriptions of scattering behaviors. Statistical approaches in such situations then often require restrictive and limiting approximations, such as the Swerling target models [3]. To improve corresponding SAR applications, it is important to perform further investigation into how radar waves are scattered in situations that elude both accurate analytical and statistical modeling.

An interferometric SAR (InSAR) uses the phase difference between a pair of SAR images acquired simultaneously (single-pass InSAR) or subsequently (repeat-pass InSAR) to measure parameters of topography and coherent temporal differences (surface displacement, atmospheric, and near surface volume-dielectric change), respectively. For the general (repeat pass) case, the interferometric phase can be expressed as a sum of six phase components

$$\varphi_1 - \varphi_2 = \varphi_{\text{flat Earth}} + \varphi_{\text{topography}} + \varphi_{\text{motion}} + \varphi_{\text{atmosphere}} + \varphi_{\text{soil moisture}} + \varphi_{\text{noise}} \mod 2\pi. \quad (1)$$

For the single-pass case, only factors one, two, and six are present. Understanding the signal and stochastic properties of InSAR data, which is governed by the second-order statistics, can add additional challenge. Some components in (1), e.g., topography, are easily expressed analytically using the acquisition geometry. However, other phase components caused by single or multiple scattering from natural (soil, snow, and vegetation) or artificial (vessels and infrastructure) near Earth surface volumes are often not fully understood [4], [5], [6], [7]. Both fundamental research and applications seeking to exploit these InSAR phase components benefit from more accurate numerical modeling of the underlying signal formation processes.

Many SAR simulators already exist and are suitable for a wide variety of purposes. Some examples include simulating SAR raw data containing urban structures on top of a rough surface [8], simulating high-resolution SAR images using modified optical ray-tracing software [9], and time-domain

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interferometric simulations [10]. Unlike the aforementioned simulators, the one presented here uses the finite-difference time-domain (FDTD) method, making it suitable for investigating combined volumetric and surface scattering of realistic near Earth surface volumes.

### B. FDTD Method

The FDTD method is a computational electrodynamics technique for simulating Maxwell's equations. The method splits space into a fine grid, over which the partial differential equations and the physical materials are discretized and solved. This type of direct numerical modeling of the physics of electromagnetic propagation and scattering has the advantage that it makes very few assumptions and is, therefore, able to simulate complex situations containing a multitude of interacting scatterers.

In numerical analysis, directly estimating derivatives using

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \tag{2}$$

with a small value of $h$ suffers from round off and truncation errors [11]. In the case of electromagnetism, this problem is alleviated by placing the field components on a Yee grid [12]. Here, each $\boldsymbol{E}$-field component is surrounded by four circulating $\boldsymbol{H}$ components, and vice versa. Using this scheme means the finite differences are computed using central differences and are second-order accurate [13].

FDTD simulations can be used to estimate scattering characteristics for situations that cannot be described by closed-form expressions. Unfortunately, because space is split into a fine mesh, FDTD simulations can be slow to run and require substantial computing power. This is further aggravated by numerical stability of the method also requiring very small time steps. The Courant–Friedrichs–Lewy condition provides an upper bound on the time step as a function of cell size and propagation velocity [13]. In addition, for stochastic evaluation of the simulated signal properties (for example of rough surface scattering), many independent simulations need to be run to get statistically significant results from a distribution of scattering elements. This increases the total simulation time even further.

Despite these challenges, FDTD has been used for modeling radar-target interactions in multiple papers [4], [14], [15], [16]. Of particular interest to our study is the 2-D FDTD simulator introduced in [4] to analyze how soil moisture and structure impact InSAR phase. This article introduces a new FDTD software package inspired by the one developed in [4]. The new software makes a number of important improvements, such as simulating three spatial dimensions and using GPU acceleration.

To perform the simulations we are interested in with an existing FDTD software package, it must support: importing user-defined full resolution grids of material values, a tightly collimated (tapered) beam field source, and high-performance GPU acceleration. Commercial FDTD tools are expensive and can be challenging to use, because we do not have access to source code to mitigate any problems found during our experimentation. Of the open-source tools, Meep [17] seems the most applicable, but it does not have GPU support. None of the other existing open-source packages, such as OpenEMS [18] or gprMax [19], have all the features we are interested in, which is why we decided to develop a specialized new FDTD simulation tool. The FDTD software described here can be used for investigating many interesting SAR targets involving the structure of snowpacks, soils, and vegetation, as well as objects buried or hidden within such scattering volumes.

## II. Simulating an SAR Resolution Cell ("Pixel")

### A. Real-Aperture Equivalent Scenario

An SAR image is formed by matched filtering the chirped pulse returns collected by a side-looking radar mounted on a moving platform [20]. The SAR image focusing process is done in both range and azimuth dimensions. In range, a chirp is transmitted, and the range response is computed using pulse compression. The theoretical resolution depends on the bandwidth of the transmitted chirp. This is equivalent to transmitting a very short pulse, which is unfeasible in practice due to limitations on peak transmit power.

In azimuth, the focusing process uses the platform motion to synthesize an aperture larger than that of the physical antenna. This synthetic aperture allows for the production of high-resolution images consisting of resolution cells (or pixels) far smaller than the area illuminated by the physical antenna. Since SAR systems typically use antennas with a wide beamwidth and transmit long chirps, directly simulating SAR raw data using a numerical method, such as FDTD, is too resource intensive to feasibly fit in computer memory. Instead, our goal is to develop a method of simulating single pulses of an equivalent real-aperture sensor (large narrow beam antenna) that would produce a ground footprint corresponding to a focused SAR resolution cell, which is much smaller than the beam footprint of the small, wide-beam physical SAR antenna. For example, the SFU SlimSAR system [21] has an L-band antenna with a 60° beamwidth, which at a slant-range of 1200 m has a footprint that spans ≈1250 m. However, the resolution of the focused L-band images is significantly smaller at approximately 0.5 m.

If we assume that the imaged scene does not change during the synthetic aperture acquisition, then a single SAR pixel will be accurately represented by modeling the equivalent real-aperture scenario. From [20, (Appendix 4C)], the synthesized antenna radiation pattern in azimuth is approximately

$$p_s(\theta) \approx L_s \operatorname{sinc}\left(\frac{2L_s}{\lambda}\theta\right) \tag{3}$$

where $L_s$ is the length of the synthetic aperture and $\lambda$ is the wavelength.

To complete our SAR analog, we replace the synthesized, real-aperture antenna located in the far-field by a near-field source located directly above the target surface (Fig. 1). This source must have a tapered shape similar to the sinc function in (3) and be excited by a temporally short pulse. Conceptually, we now have an FDTD simulation space small enough to fit into computer memory when discretized into Yee cells and an understanding of the incident beam shape required.

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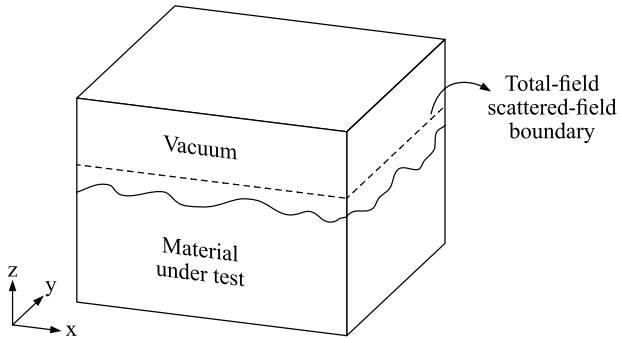



Fig. 1. Layout of the FDTD simulation space. The top of the simulation grid is vacuum, and a tapered incident beam is fed into the simulation space at the TF/SF boundary. Each face of the simulation space is convolutional perfectly matched layer absorbers.

Using a tapered incident beam serves two purposes. The first, described above, is to create a real-aperture analogy to a focused SAR pixel. The second purpose is to enable rough surfaces to be simulated without diffraction at the edge of the simulation space. If the beam footprint is chosen appropriately, the incident field will be close to planar in the center of the simulation space and taper down to a negligible amount near the edges. Since the fields are null where the rough surface meets the simulation boundaries, there is no edge diffraction. This is the technique used in this software package and was successfully demonstrated previously in 2-D by [4]. Any remaining reflections from the simulation boundaries are avoided by using convolutional perfectly matched layer absorbers [13].

Rather than using a beam with a sinc weighting as described in (3), we use a Gaussian weighting to avoid sidelobes. It is possible to emulate data from an SAR sensor by choosing the width of the Gaussian amplitude profile to be similar to the width of the SAR's impulse response. Furthermore, low-resolution sensors can be emulated by simulating higher resolution data (i.e., using a smaller simulation space) and then coherently summing the response from multiple simulations.

### B. Incident Beam Generation

The short pulse for the SAR equivalent real-aperture scenario is constructed based on the tapered beam method described in [22], which we chose because it has a Gaussian amplitude profile, and according to the author is "*reliably tapered and clearly polarized for all angles of incidence*." This beam was specifically designed for investigating scattering from rough surfaces, but with the method-of-moments technique, not FDTD. It, therefore, defines the beam based on a single, real wavenumber $k$, which is the wavenumber of the central plane wave of the beam.

However, our FDTD model of an SAR real-aperture equivalent scenario must be excited by a time-domain pulse with a predefined bandwidth. The shape of the pulse is a Gaussian-weighted sine. Using the dispersion relation, $k = (\omega/c)$, we can find a wavenumber for each angular frequency $\omega$ in the excitation pulse, and the weight of each $\omega$ can be found using a discrete Fourier transform. Then, the method in [22] can be applied to each $k$ separately and the result weighted by the Fourier transform coefficient for frequency $\omega$. This can then be transformed back to the time domain by doing an inverse Fourier transform (Fig. 2). The resulting short-pulsed beam is a solution of Maxwell's equations, because it is a spatial and temporal superposition of plane waves.

The field values constructed as described above are then used as the excitation for an incident FDTD simulation. In this simulation, field values across an $xy$ plane are recorded. These recorded fields are fed into the main simulations using the total-field/scattered-field (TF/SF) approach [13]. This technique allows the incident beam to be injected at a specific plane in the simulation space, while allowing the scattered fields to pass through unperturbed. The procedure requires knowing the tangential components of $\boldsymbol{E}$ and $\boldsymbol{H}$ along the source plane for each point in time that the excitation pulse is active. For our simulation setup (Fig. 1), this means the TF/SF boundary is located on an $xy$ plane placed just above the surface of the material under test.

The lower bound on the size of a simulation space is constrained by the minimum width of the tapered incident beam. The width of the incident beam (defined as the radius at which the amplitude falls to $1/e$ of its maximum) should be greater than $2\lambda$ to minimize errors [22]. This is smaller than a typical SAR resolution cell, and most simulation spaces will be above this minimum. The upper bound on the simulation size is limited by computer memory.

### C. Comparison With Tapered Beam in 2-D

The beam setup described above is more complicated than the incident pulsed beam used in the 2-D simulation described in [4]. This is required for the fields to remain valid solutions of Maxwell's equations in three dimensions.

*1) Two-Dimensional Method:* In 2-D, the simulation space in [4] existed in the $XZ$ plane. A tapered beam is simulated traveling straight down (in negative $z$) in parallel to the main simulation. At each time step, the fields in the incident simulation are interpolated along a line defined by the incident angle to compute the field values fed into the TF/SF boundary in the main simulation. For both linear polarizations (horizontal and vertical), there is a field component that is orthogonal to the beam and grid ($E_y$ for horizontal and $H_y$ for vertical). The excitation to the incident simulation with horizontal polarization is a Gaussian-weighted sinusoid in time with a Gaussian spatial footprint. This is expressed as follows:

$$E_y(x,t) = A\exp\left(-\frac{t^2}{\sigma_t^2} - \frac{x^2}{\sigma_x^2}\right)\sin(\omega t). \tag{4}$$

There is no variation of the field values with respect to y in (4). This means the divergence of the electric field is

$$\nabla\cdot\boldsymbol{E} = \underbrace{\frac{\partial E_x}{\partial x}}_{=0} + \underbrace{\frac{\partial E_y}{\partial y}}_{=0} + \underbrace{\frac{\partial E_z}{\partial z}}_{=0} \tag{5}$$

which is always zero irrespective of the frequency.

*2) Three-Dimensional Method:* In 3-D, the fields are constructed as a superposition of plane waves [22]

$$\boldsymbol{E}(\boldsymbol{r}) = \int_{-\infty}^{\infty} d\boldsymbol{k}_\rho \exp(j\boldsymbol{k}_\rho\cdot\boldsymbol{\rho} - k_z z)\psi(\boldsymbol{k}_\rho)\boldsymbol{e}(\boldsymbol{k}_\rho) \tag{6}$$

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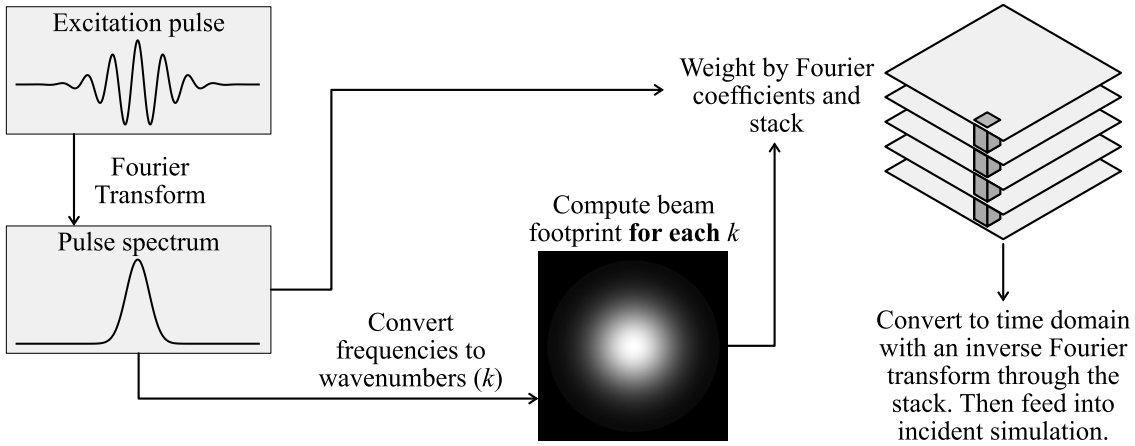



Fig. 2. Excitation to the incident FDTD simulation is generated by superimposing plane waves spatially and temporally. The excitation pulse is a Gaussian-weighted sine, and the beam footprint is described in [22].

where $\boldsymbol{r} = \boldsymbol{\rho} + \hat{z}z$ and $\boldsymbol{k}_\rho = \hat{x}k_x + \hat{y}k_y$. Next, $\psi(\boldsymbol{k}_\rho)$ defines the beam footprint (a Gaussian envelope in our case), and $\boldsymbol{e}(\boldsymbol{k}_\rho)$ defines the polarization. If the polarization vector $\boldsymbol{e}(\boldsymbol{k}_\rho)$ is constant, then $\boldsymbol{E}(\boldsymbol{r})$ is not a valid solution of Maxwell's equations, because, in general, $\nabla \cdot \boldsymbol{E}(\boldsymbol{r}) \neq 0$ [22]. Creating a polarization vector, such that $\nabla \cdot \boldsymbol{E}(\boldsymbol{r}) = 0$, introduces a dependency on the central wavenumber of the beam. Therefore, this field is a valid solution of Maxwell's equations for only a single frequency. This means that to construct a valid field, the tapered beam field along the $xy$ plane $\boldsymbol{E}(\boldsymbol{r})$ must be computed for each frequency component in the temporal excitation pulse and then linearly superimposed, as shown in Fig. 2.

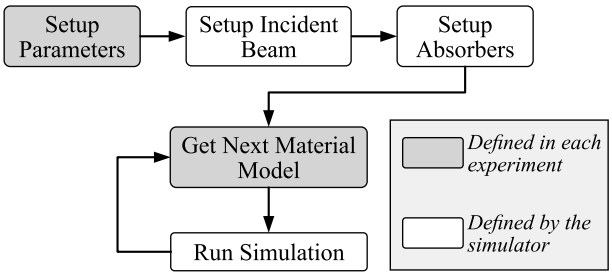



Fig. 3. Flowchart displaying how a series of FDTD simulations are run to complete an experiment. The parameter setup and material grid creation components are user-defined for each experiment. The other components are common to all experiments and are defined by the FDTD simulator software package.

## III. Software Design

### A. Design Goals

The software design is focused around two goals.

1) Allow the user of the FDTD simulator to define the input material grids, and how they are generated.
2) Operate fast enough to perform repeated simulations while varying parameters and gathering backscatter statistics.

The first goal is met by splitting the software into clearly defined modules. Some modules are defined by the user before running an experiment, and others are defined by the FDTD simulator. This separation is shown in Fig. 3.

The second goal is more challenging to achieve. Realizing fast performance has been approached in three ways: 1) use different implementations to target specific hardware, such as GPUs or FGPAs; 2) run independent simulations concurrently; and 3) reuse pregenerated incident beams whenever possible. Currently, the supported processing hardware is CPUs and NVIDIA GPUs.

The software was written in Python, with the GPU kernels written in CUDA C. Python was chosen due to its ubiquitous nature in scientific computing as well as its large environment of libraries. No performance critical functions are executed by the Python interpreter, which itself is not suitable for high-speed numerical computations. We use two libraries to achieve our performance goals. First, the CPU simulations use Numba [23], which is a JIT compiler based on the LLVM compiler infrastructure. Second, the GPU simulations are run using PyCUDA [24]. This library provides a simple interface between GPU and CPU memory, as well as a way to compile and invoke CUDA kernels.

Table I compares simulation times across different hardware platforms for a typical simulation size and duration. The importance of using GPU's to accelerate the simulations is clear, as the GPU simulation times are approximately 10–15 times faster than a single-threaded CPU implementation. This table only illustrates typical simulation times, as the exact timing is sensitive to both computer hardware and simulation parameters.

### B. Memory Usage Analysis

Memory consumption is the key limiting factor to the maximum size of each simulation, especially when using GPUs. High-end modern GPUs, such as the NVIDIA V100, only have 32 GB of memory, which is a relatively small amount compared with servers, which can easily have hundreds of gigabytes. The two main factors impacting memory usage are the total volume of the simulation space and the grid density. The plots in Fig. 4 show how memory usage changes with FDTD grid spacing and simulation space volume. Important

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TABLE I
COMPARING SIMULATION TIMES ACROSS DIFFERENT COMPUTING PLATFORMS

| **Common Parameters** | | |
|---|---|---|
| Number of cells | $81.3 \times 10^6$ | |
| Number of timesteps | 1440 | |
| **Computing Device** | **Numerical Precision** | **Time** |
| Nvidia Tesla K80 | 32-bit | 122 s |
| Nvidia Tesla K80 | 64-bit | 229 s |
| Nvidia RTX 3060 | 32-bit | 73 s |
| Intel Xeon Gold 6346 | 32-bit | 1057 s |
| Intel Xeon Gold 6346 | 64-bit | 1802 s |

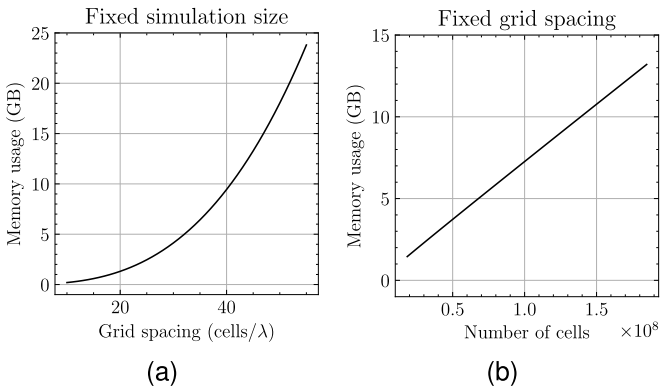



Fig. 4. Plots showing how simulation size and grid spacing impact memory usage. (a) Memory usage versus grid spacing for a fixed simulation size of $16\lambda \times 16\lambda \times 8\lambda$. (b) Memory usage versus simulation volume for a fixed grid spacing of 40 cells/$\lambda$.

to note is that memory usage increases linearly with simulation space volume, but increases with the cube of the grid spacing. Since the grid spacing is a constant (no adaptive meshing), material models containing higher permittivity materials require a finer mesh for the entire simulation space due to wavelength compression in material by index of refraction. This is an important parameter to consider when designing experiments.

### C. Running Concurrent Simulations

This simulator routinely runs experiments, which consist of repeatedly injecting the same incident beam into many randomly generated soil models. The individual results from each soil model realization are independent of the others, which means they can be run concurrently in any order.

This functionality is implemented using a manager–worker scheme (Fig. 5). One node is given the role of manager. The manager creates new material models and distributes them to idle workers. Once workers have completed a simulation, they inform the manager they are idle, and wait to be given another material model to simulate. The manager and workers communicate using MPI, which allows this solution to scale from PCs to clusters.

## IV. SIMULATOR VALIDATION

A key weakness of the FDTD method as compared with an analytical model, such as [25], is that the software must

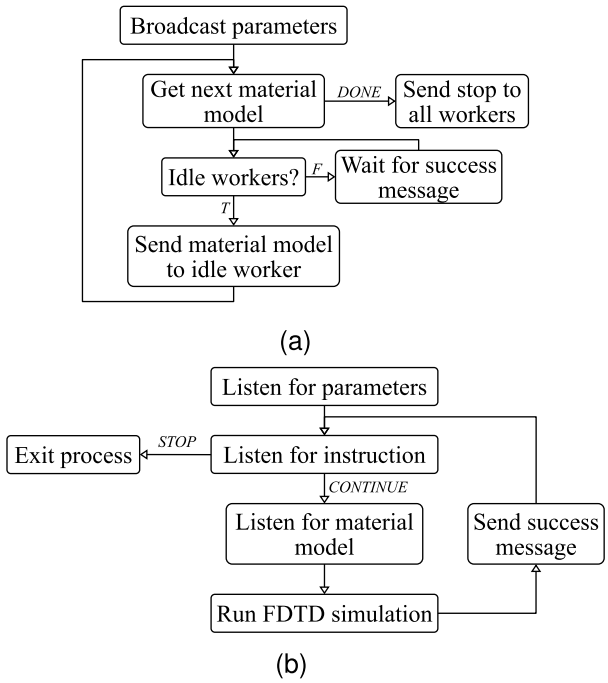



Fig. 5. Flowcharts showing how the manager and worker processes interact while running concurrent simulations. (a) Manager process. (b) Worker processes.

be carefully validated before results can be fully trusted. The validation process for our new software started with simple, qualitative observations and progressed to quantitative comparisons with known theoretical results.

The first validation step was to perform qualitative inspections of the $\boldsymbol{E}$ and $\boldsymbol{H}$ fields during simulations. These inspections were done continuously during development. Their purpose was to quickly find software bugs, such as indexing errors, that cause the fields to immediately behave erratically. Obvious problems were quickly identified and fixed before the following quantitative evaluations were performed.

In the first quantitative test, we injected the beam into a simulation space consisting solely of vacuum for the four combinations of GPU/CPU processing target and single/double-precision floating point values. During each of the simulations, phasors were recorded over the interior six faces of the simulation space. Each of the results were in close agreement with each other. This test is important to ensure the four operational modes of the software produce equivalent results.

The next validation step ensures the simulations are not erroneously gaining or losing energy. Similar to the previous test, an incident beam is injected into a vacuum. Phasors at the frequency of interest are recorded along the incident beam plane and the interior faces of the simulation space. Subsequently, the Poynting vector is computed at each cell location using the phasors. The total power in and out can then be compared by performing surface integrals of the Poynting vectors. The difference between input and output power is less than 0.001%, which indicates that the simulation space is conserving energy.

Following this, we compare the reflected power measured during simulations against theoretical solutions. The goal of

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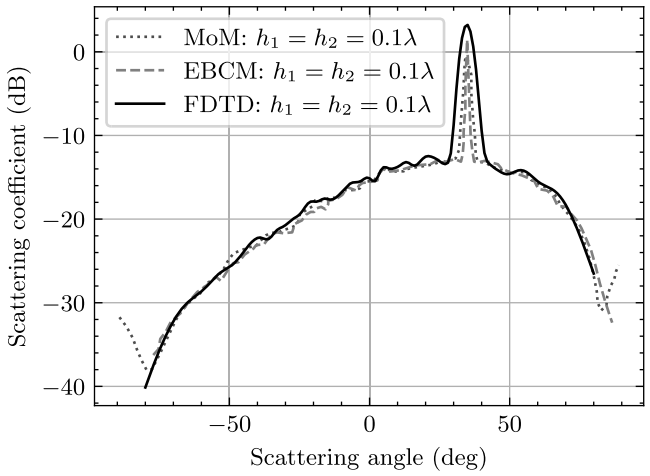



Fig. 6. Comparing the bistatic scattering coefficient computed with the FDTD method presented here to results from two other methods published in [27]. The material under test is a two-layer rough dielectric. This validation test is identical to one performed on the 2-D FDTD simulator in [4].

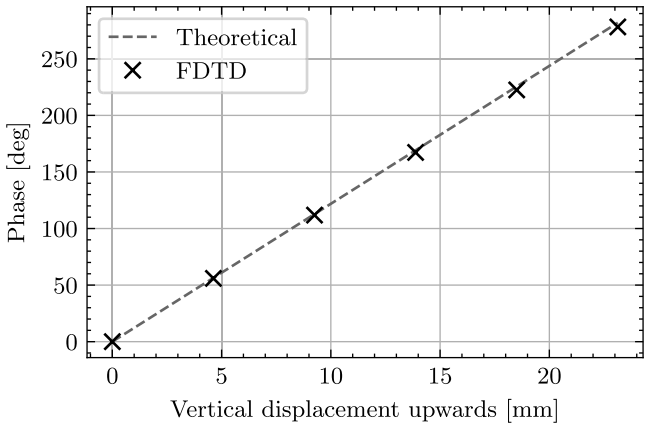



Fig. 7. Comparing simulated InSAR phase against the theoretical values. A collection of homogeneous rough-capped dielectric slabs were simulated at different vertical heights. The theoretical phase can be computed using the geometric path length change. Simulation parameters are as follows: $f = 5.4$ GHz, $\theta_i = 20°$, $h_{\text{rms}} = 5$ mm, and $L_{\text{corr}} = 2$ cm.

these tests is to establish reasonable simulation parameters for the experiments in Section V and understand the expected error magnitude. The FDTD cell size was chosen, such that there were at least 15 cells per wavelength in the material under test, which balanced simulation accuracy and resource usage. The first test is done using a perfectly smooth dielectric half-space with a relative permittivity corresponding to the soil moisture values investigated in Section V-B. The relative permittivity of the dielectric was computed using the Hallikainen model [26] with identical soil parameters as in Section V-B, so that realistic performance estimates could be gathered for this use case. Multiple simulations were done with different parameter configurations to evaluate their impact on the error. This included changing the incident angle, simulation space size, simulation duration, floating point precision, and cell size. From this, we found that the reflected power in the simulations had a maximum 0.11-dB deviation from the theoretical solution. After this, we simulated a two-layer dielectric with perfectly flat interfaces. The top layer had the properties of dry soil, and the bottom layer had the properties of wet soil. The maximum error in this case was 0.35 dB for both HH and VV polarizations. This is similar to the results from the two-layer validation test performed on the 2-D FDTD simulator in [4], which indicates the 3-D FDTD tool presented here should produce similar results.

The next validation test compares rough surface scattering against previously published results of two other methods—the method of moments (MoMs) and the extended boundary condition method (EBCM). The example was taken from [27, Fig. 10(b) and Table III], and consists of two uncorrelated rough layers separated by $2\lambda$ with a 35° incident angle. Since the MoM and EBCM results were computed in 2-D and the FDTD simulator uses a 3-D space, we use a grooved surface (only rough in one dimension) rather than a 2-D isotropically rough surface. The comparison among FDTD, MoM, and EBCM is shown in Fig. 6. Each of the three methods closely agree for the majority of scattering angles. The forward scattering peak is wider in the FDTD results than the other two methods, which is due to the FDTD simulation space being approximately 80% of the size used in [27]. This test also confirms that 2-D results can be recreated with the 3-D simulator by creating materials that do not vary in the third dimension.

Finally, we perform a validation test for the InSAR phase. A collection of 32 rough surfaces were generated with homogeneous material properties (no volume scattering). These surfaces were generated at a number of different mean heights, and the response at each height was multilooked together. This multilooked phase closely follows the path length change, as shown in Fig. 7. The ideal theoretical phase is given by

$$\varphi_{\text{ideal}} = \frac{4\pi\ \Delta r}{\lambda} = \frac{4\pi\ \Delta h \cos\theta}{\lambda} \tag{7}$$

where $\Delta r$ is the path length change, $\Delta h$ is the vertical surface displacement, and $\theta$ is the incident angle. The phase increases as the surface moves closer to the sensor, which is the same convention as used in [4].

## V. Demonstration of Capabilities

### A. SAR Data Simulation

One of the simplest uses of the FDTD simulator is to simulate a small SAR scene pixel by pixel. This can be useful for a qualitative analysis in which a visual representation is more important than scattering statistics.

An example scene containing two bare fields separated by a dirt road is shown in Fig. 8. The simulated SAR scene was generated in the following way. First, the surface profile and material properties were generated for the entire scene. The scene is then split into pixel sized chunks and fed into the simulator. After all the simulations are complete, the backscatter amplitude values are arranged into an image.

### B. InSAR Phase Simulation

A major advantage of simulating SAR data using a numerical method is that it is possible to construct scenarios that

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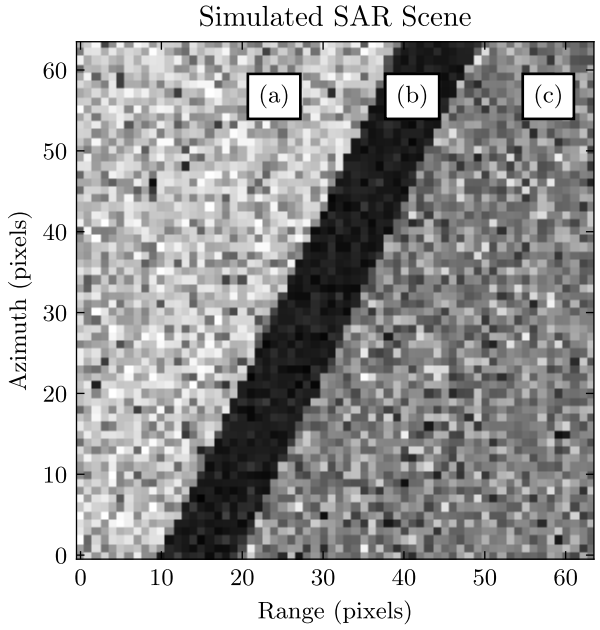



Fig. 8. Simulated 64 × 64 pixel C-band SAR image with three regions. Each pixel is 0.9 × 0.9 m. The three regions consist of a homogeneous soil with the properties: (a) $h_{\text{rms}} = 10$ mm, $L_{\text{corr}} = 2$ cm, and SM = 15%, (b) $h_{\text{rms}} = 5$ mm, $L_{\text{corr}} = 15$ cm, and SM = 5%, and (c) $h_{\text{rms}} = 7$ mm, $L_{\text{corr}} = 6$ cm, and SM = 10%.

are difficult to solve analytically. Statistics on the scattering characteristics can be gathered by generating many realizations of the rough surfaces. The FDTD simulations can also be used for InSAR, because the recorded phasors have both amplitude and phase information.

The researchers in [4] used similar 2-D FDTD simulations to investigate the impact of soil moisture and structure on InSAR phase. In this experiment, we recreate some of their results to show continuity between the two tools. To mimic the 2-D results, a rough surface profile containing grooves running in the $Y$-axis is used, rather than an isotropically rough surface. The InSAR phase reported is the difference phase, where the phase of the driest soil in each experiment has been subtracted. This allows us to observe the relative phase change as a function of moisture for each soil type experiment.

The scenario is set up as follows. The incident angle, surface roughness, layer thickness, and interlayer correlation parameters were chosen to be identical to the vertical moisture gradient experiment in [4]. The soil is 95.5% sand, and the permittivity and conductivity as a function of moisture are computed using the Hallikainen model [26]. Rather than the homogeneous moisture profile used above in Section V-A, this experiment uses a vertical gradient profile. The very top of the soil is 3% moisture by volume, while the bottom ranges from 3% to 30% with a smooth transition between the two.

The phase and amplitude are averaged over 64 simulations for each soil moisture value to average out speckle. This is equivalent to multilooking over an 8 × 8 window. For each of the soil moisture values, the 64 rough surfaces are identical—the only thing that changes between acquisitions is the moisture, not the surface realizations.

The results for both a grooved "2-D equivalent" and a "fully 3-D" isotropic rough surface are shown in Fig. 9. The results

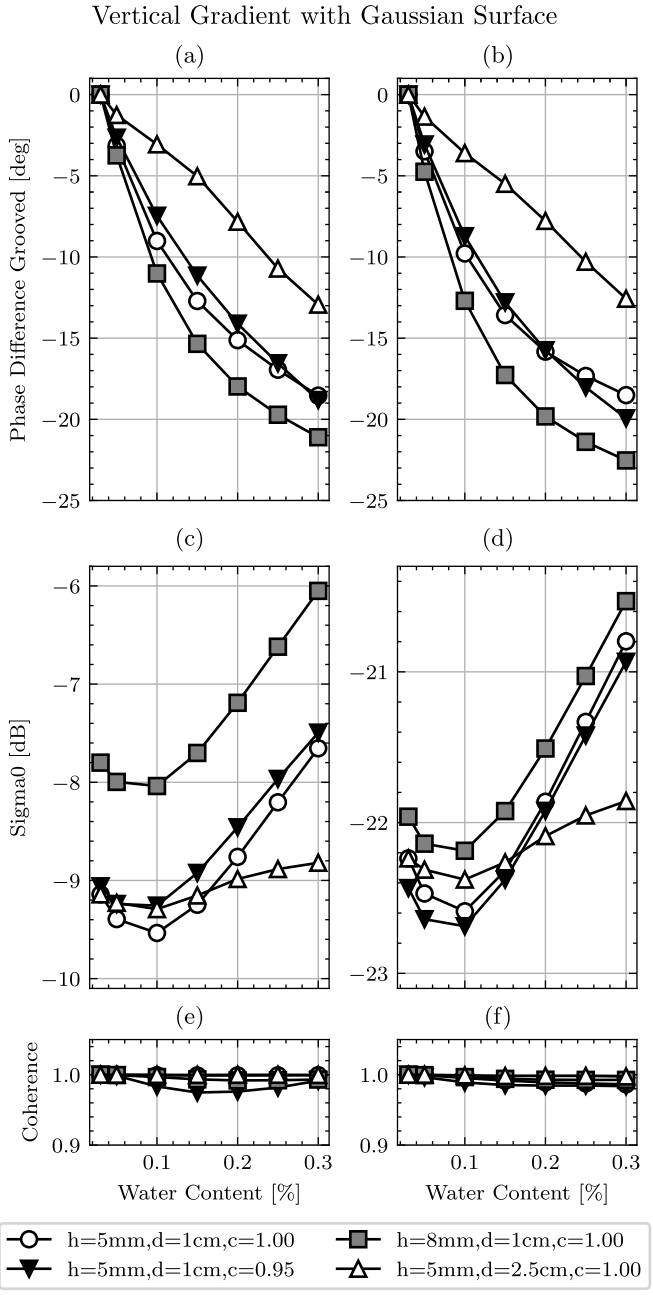



Fig. 9. Scattering characteristics for HH polarization incident to soil with a Gaussian rough surface and vertical gradient moisture profile. The left column contains results from a "2-D equivalent" grooved surface, and the right contains results from an isotropically rough surface. Parameters are as follows: $f = 5.4$ GHz, $L_{\text{corr}} = 2$ cm, and $\theta_i = 20°$. (a) and (b) Interferometric phase. (c) and (d) Backscatter. (e) and (f) Coherence magnitude.

from the 2-D equivalent surface are very similar to those in [4, Fig. 7] despite the differences between the two simulation tools. There is a small dip in backscatter due to resonance between the two layers. As expected, the backscatter in the isotropic rough surface experiment is less than the grooved experiment, and in this case, the decrease is approximately 14 dB. This is because the grooved surface will reflect the incident beam only forward and backward, but not sideways. The bumps on the isotropic rough surface will scatter in all directions, so comparatively less energy reflects toward the radar sensor.

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As well, the phase response is significant for both surface types, with a magnitude up to approximately 20° for both grooved and isotropic cases. Interestingly, the phase response is very similar between the grooved and isotropic soil surfaces. This indicates that in this particular case, the InSAR phase is not overly sensitive to the surface roughness.

Importantly, the coherence for all four parameter sets is high (greater than 0.95), which means that changing the interlayer spacing does not destroy InSAR coherence. From this, we can draw the same conclusions as in Section IV-D of [4], namely, that phase changes from vertical moisture gradients are readily observable with spaceborne InSAR. Compared with when [4] was published, this is no longer a novel discovery, but is important, because it shows that the FDTD simulator presented in this article can reproduce results consistent with those of other tools and models.

### C. Detecting Buried Objects

Characterizing the scattering from buried targets is challenging, but of great value to remote-sensing practitioners. For example, the use of SAR to detect landmines from UAVs [28] would greatly reduce risk in demining. Therefore, after completing validation as our first demonstration experiment, we show how FDTD simulations can predict the backscatter strength of buried mines modeled as simple cylinders of different materials embedded in moist soil.

An isotropically rough soil model is generated with the vertical moisture gradient used in Section V-B. The top of the soil has 3% moisture by volume, while the bottom varies from 3% to 30% with a smooth transition between the two over a 1-cm layer thickness. The cylinders are 30 cm in diameter, 10-cm tall, and are positioned upright with the top surface placed 8 cm below the mean surface height of the soil.

Three different scenarios are simulated, each with 64 different rough surface realizations, and the results averaged to remove speckle (Fig. 10). The first scenario is when no cylinder is present in the soil, followed by a cylinder made out of plastic (high-density polyethylene with dielectric properties computed using [29]), and last a metal cylinder.

As expected, for most moisture values, the metal cylinder causes the highest backscatter. The backscatter for the metal cylinder peaks at more than 8 dB larger than without buried target. At the highest soil moisture, all three scenarios have almost the same backscatter. This is because the path loss in moist soil is much higher than in dry soil, which serves to obscure the object placed in the soil. Our future plan is to use the FDTD simulator to help develop a robust mine detection application for SAR through systematic characterization of how combinations of soil and buried object properties impact backscatter. The simulator can both produce diverse test data to train detection algorithms and quantify operational performance bounds.

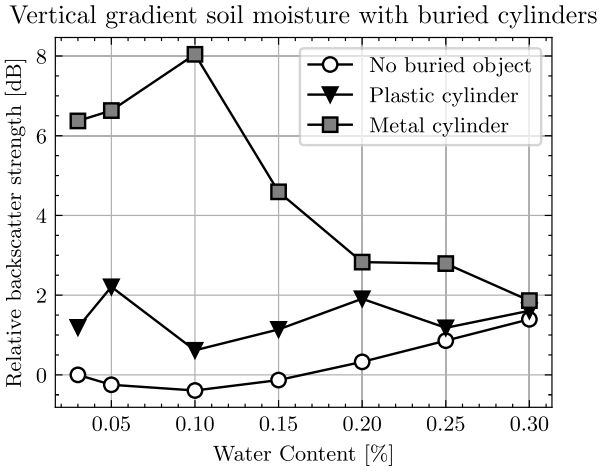



Fig. 10. Comparing backscatter strength for three different scenarios: no buried object, a buried HDPE cylinder, and a buried metal cylinder. HH polarization incident to soil with a Gaussian rough surface and vertical gradient moisture profile. Parameters are as follows: $f = 5.4$ GHz, $\theta_i = 20°$, $h = 5$ mm, $L_{\text{corr}} = 2$ cm, and $d = 1$ cm.

## VI. Discussion

### A. Surface Waves

In [4], it was noted that the fields near the soil surface are stronger when a vertically polarized incident beam is used, as compared with horizontal polarization. The same qualitative behavior is also seen in our 3-D simulation (Fig. 11). While, in [4], Zenneck waves [30] were suspected as a possible explanation, this has since been ruled out as per the following argument. Zenneck waves are linked to the Brewster angle [31]. In the cases we are interested in, the incident wave travels through vacuum into a denser medium ($\varepsilon_r > 1$). The Brewster angle is $\theta_B = \arctan(n_2/n_1)$, and since $n_2 > n_1$ in our case, the Brewster angle is always greater than 45°. The experiments in [4] use an incident angle of 20°, which is smaller than the Brewster angle for the soils under investigation. This implies the observed phenomenon cannot be a Zenneck wave. Our FDTD tool in its current form directly reveals the absence of sideways energy loss in this case; the tool is well suited to reveal guided surface wave phenomena if they occur.

Rather, the greater power observed near the surface for vertical polarization can be explained by analyzing the boundary conditions of a simplified situation—a plane wave incident to a lossy dielectric half-space. In this case, we can compute the power transmitted through the vacuum-dielectric interface with Fresnel power coefficients. Using the soil parameters under investigation in Section V-B, the transmission and reflection coefficients for both vertical and horizontal polarizations are shown in Fig. 12. For every moisture value, the vertical transmission coefficient is greater than that of the horizontal polarization. This corroborates our observations with the 3-D FDTD simulator described here (as well as retroactively the analogous phenomenon observed in Fig. 3 of [4]). One implication of this result is that users of our simulator software must take into account the polarization of interest when deciding how large the simulation space should be relative to the width of the incident beam. This is illustrated in Fig. 11, which shows that the VV polarized incident beam has a larger power level in the soil. To obtain accurate results, ideally, all the scattering should occur within the simulation space. This can be inspected after running a simulation by comparing the

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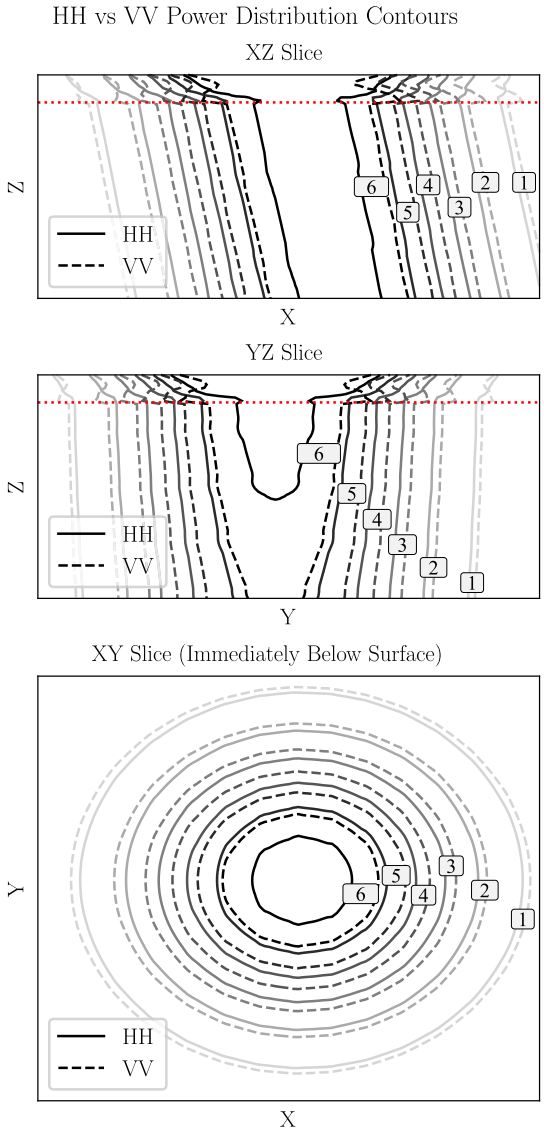



Fig. 11. Comparison of the power distribution throughout the 3-D simulation space for HH and VV polarizations. This was computed using the temporal mean of the Poynting vector amplitude. The material is a flat 95.5% sand soil with 10% moisture, $f = 5.4$ GHz and $\theta_{\text{inc}} = 20°$. The red dotted line marks the position of the soil surface, while the numbered boxes overlap matching pairs of contour lines between the two polarizations. As predicted, the vertical polarized beam extends wider into the soil than the horizontal polarization.

power leaving the sides of soil to the power in the incident beam.

### B. Limitations

The main limiting assumption made by this tool is that the synthetic aperture can be replaced by a real-aperture equivalent. This enables us to apply the FDTD method by allowing us to construct a simulation grid small enough to fit into computer memory. The largest implication from this choice is that any changes, which occur during the synthetic aperture, cannot be modeled. For example, this simulator would be a poor tool for investigating how moving ships or vehicles appear in SAR images.

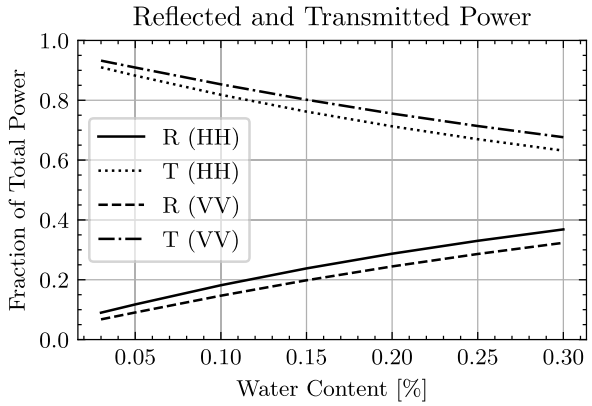



Fig. 12. Power reflection and transmission coefficients for an interface between vacuum and a 95.5% sand soil for $f = 5.4$ GHz and $\theta_{\text{inc}} = 20°$. The horizontal (TE) polarization has a smaller transmission than the vertical (TM) polarization.

TABLE II
COMPARISON OF PENETRATION DEPTHS FOR VARIOUS MATERIALS AT THE C-BAND

| Frequency | Material | Penetration Depth |
|---|---|---|
| 5.4 GHz | Soil: 95% sand, 5% water | 0.082 m |
| 5.4 GHz | Soil: 95% sand, 10% water | 0.038 m |
| 5.4 GHz | Soil: 95% sand, 20% water | 0.021 m |
| 6.0 GHz | Dry snow: 0.05% water | 69.7 m |
| 6.0 GHz | Moist snow: 1% water | 1.67 m |
| 6.0 GHz | Wet snow: 7% water | 0.085 m |

The process of replacing the far-field real-aperture equivalent with a near-field source assumes that the intervening space is vacuum. This means atmospheric effects, such as attenuation, dispersion, and phase delay, are all excluded from the results. The near-to-far-field transform performed at the end of a simulation also assumes that the recorded near-field data settle down to zero. However, while this requirement may put a lower limit on simulation time, it can be confirmed to be met during processing.

As well, if the relaxation time [32], $\tau = (\varepsilon/4\pi\sigma)$, of the material under test is long enough to span multiple transmitted pulses, this could impact the scattering in a way that cannot be modeled with this tool. However, effects related to the relaxation time are unlikely to have any appreciable impact. Even for a very low-loss (i.e., low conductivity) material, such as dry snow at the L-band [33], the relaxation time is on the order of $10^{-8}$ s, which is far smaller than the target illumination time in a typical SAR.

The simulator is also difficult to use in scenarios with deep radar penetration. This is not for a theoretical reason, but because the size of the simulation space is limited by computer/GPU memory. Moist soils are an ideal simulation target, because the wave penetrates into the soil, but not as deeply as in dry snow or dry sand. The penetration depth (without accounting for any volume scattering) can be quantified as the distance it takes for the power to drop by a factor of $1/e$ compared with the surface. The values of penetration depth for a few different materials [26], [33] are shown in Table II.

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Total simulation times are also greatly increased by the need to capture scattering statistics. A single simulation run only provides the results on one particular rough surface realization. However, we are usually interested in statistics, such as the mean backscatter power, for a certain roughness profile, or InSAR coherence over a time series. The total simulation time to obtain results then linearly increases with the number of simulations required. The number of simulations depends on the desired accuracy of the estimations and the materials under investigation. For example, in the case of distributed scatterers, the Cramer–Rao bound on the InSAR phase standard deviation as a function of number of looks is given in [34] as follows:

$$\sqrt{\left\langle \left(\hat{\Phi} - \left\langle \hat{\Phi} \right\rangle\right)^2 \right\rangle} = \frac{1}{\sqrt{2N_L}} \frac{\sqrt{1-\gamma^2}}{\gamma} \tag{8}$$

where $N_L$ is the number of looks (equivalent to the number of simulations). Therefore, high-coherence scenarios require fewer simulations than low-coherence scenarios to achieve an accurate estimation of InSAR phase. Achieving a phase standard deviation of 5° requires 197 looks with a coherence magnitude of 0.5, but only 37 looks if the coherence magnitude is 0.8. Assuming that each simulation takes 5 min on a GPU (this is consistent with the simulations in Section V), 197 looks for an InSAR pair would require 32.8 h of simulation time, while 37 looks would only require 6.2 h.

### C. Future Plans

The tool presented here can be used to investigate scattering characteristics of diverse scenarios. A key area of research it will be used in is improving soil moisture estimation from SAR/InSAR data. Analytical scattering models use assumptions about the scenario to make the problem under investigation tractable. These assumptions can include rough surface profile parameters, moisture distribution, or where scattering occurs (surface or volume). Therefore, using these models requires careful evaluation of their validity bounds. For instance, Mira et al. [35] use the scattering model developed by De Zan et al. [25] in their algorithm, which uses the first-order Born approximation with a uniform vertical scattering profile. However, a fast and accurate simulator could propel the development of more accurate soil moisture estimation algorithms. The parameter space is large (various soil structures, roughness profiles, and moisture values), but with enough simulation runs, the joint distribution mapping soil parameters to backscatter statistics could be adequately sampled. The resulting database would replace nonexistent or expensive to create real measurements. Then, a machine learning method could be trained on the simulated dataset and learn how to estimate soil moisture based on measured backscatter/phase.

Another improvement to the soil simulations will be to add vegetation, since soil moisture measurements are typically performed on land for agriculture or range land, which is not bare. Vegetation contains a variable moisture content and impacts both backscatter as well as interferometric coherence [36]. The spatial structure of vegetation varies with species and life-cycle stages, so the impact on polarimetric data can be explored as well. Linking together soil moisture, soil structure and crop state together in one simulation would be very powerful.

The simulator can also be used to investigate more esoteric SAR applications. For example, SAR has been proposed as a method for detecting polar bear dens buried under a thick layer of snow [37]. Collecting SAR data containing known bear dens is possible, but challenging. In these types of situations, simulations can be used to bolster limited datasets for detection/classification algorithm development.

## VII. Conclusion

This article presents a novel tool for investigating the scattering characteristics of individual SAR pixels. It uses a 3-D implementation of the FDTD method to simulate a beam incident upon a medium specified by the user. Rather than simulating the entire synthetic aperture, the tool uses a real-aperture equivalent, which allows us to simulate pixels individually. The largest drawback of this strategy is that dynamic scenes with changes happening within the synthetic aperture time cannot be modeled. We validated the software by comparing results with known analytical solutions, as well as comparing against other rough surface scattering methods. Following this, as first tests, we demonstrate some example uses, which include generating a simulated SAR image, recreating the vertical moisture gradient experiment from [4] and analyzing how buried objects impact backscatter as a function of soil moisture. This tool has been shown to be a capable method of investigating scattering characteristics and can be used to improve SAR and InSAR applications in the future.

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